GEK-91333A



GE Drive Systems

INSTRUCTIONS

GUIDANCE SERVICE

"SYSTEM INTERCONNECTION"

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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1. INTRODUCTION

The elementary diagram format has been drawn in such a manner as to allow easy identification of wires to be connected by the purchaser between all control equipment furnished by GE and associated remote mounted components.

Reliable installation information may be obtained directly from outlines, elementaries, panel layouts and terminal board arrangements.

2. SERVICE DRAWING DEFINITIONS

2.1 DRAWING SCHEDULES

The drawing schedules obtained from GE Drives Products Operation designates drawings as **"PRELIMINARY", "FOR INSTALLATION (AS RELEASED FOR MANUFACTURE)"**, and, **"RECORD CHANGES (AS MANUFACTURED AND TESTED)"**.

PRELIMINARY

This transmittal consists of the sketched elementary diagrams that identify terminals, wire numbers, wiring level designations, and defines wire runs between control equipment and to other areas Outline information is included when required to supplement the elementary. This transmittal enables a customer to begin a preliminary conduit and cable schedule. A transmittal letter accompanying the drawings defines them as "PRELIMINARY". Preliminary drawings are provided only upon direct request.

FOR INSTALLATION (OR AS RELEASED FOR MANUFACTURE)

This transmittal consists of the elementaries with the terminal outputs and wiring level designations identified to define wire runs between control equipment and to other areas. Outline and panel layout drawings are provided showing final equipment dimensions, conduit space available, electrical and mechanical restrictions, and exact terminal board locations. These drawings provide complete information for a final cable and conduit schedule. A transmittal letter accompanying the drawings defines them as "FOR INSTALLATION".

RECORD CHANGES (OR AS MANUFACTURED AND TESTED)

This transmittal consists of only those sheets of elementaries and layouts that changes have been made to since the "AS RELEASED FOR MANUFACTURE" transmittal. Also included are any revised terminal board or outline drawings. A transmittal letter accompanying the drawings defines them as "RECORD CHANGES".

2.2 DRAWINGS PROVIDED

ELEMENTARY

This drawing is an electrical representation of the equipment being supplied showing both the internal and external connections required to complete the electrical system. This interconnection instruction explains, by examples, the use of the elementary drawing to identify the wiring by the customer between equipment supplied by GE and to other areas.

OUTLINE

This drawing is a physical representation of the equipment to be furnished. **Preliminary** outlines provide basic overall dimensions, while **final** outlines show specific dimensions. Both contain sufficient information to allow planning of physical locations of equipment.

PANEL LAYOUT

These drawings show the exact locations of interconnecting points, and are to assist with cable routing. Terminal board and component arrangement is an integral part of the final drawing information. For planning purposes, assume the panel device location as defined by the elementaries is within the confines of the panel or panels to which the drive numbers are assigned.

OPERATOR STATION ARRANGEMENT

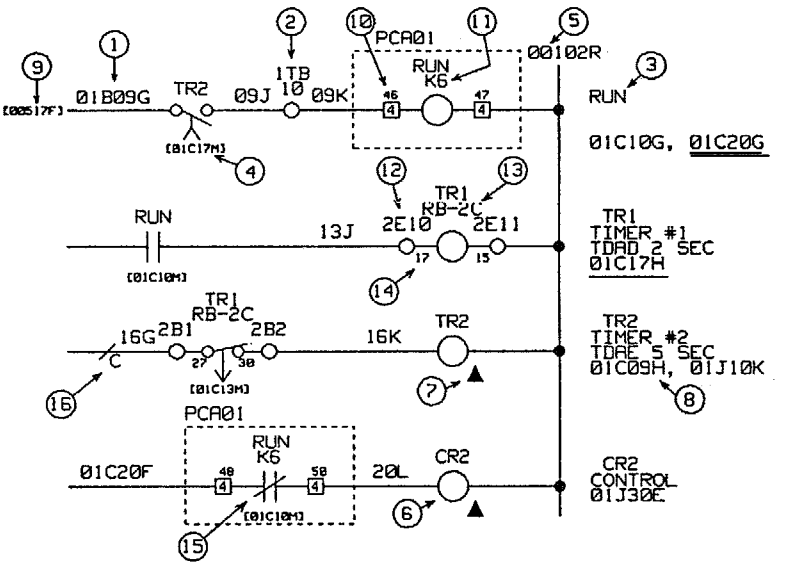
This drawing is a pictorial arrangement of the operator station devices.

3. DRAWING NOTES

3.1 SYMBOL AND NUMBER INTERPRETATION

The following drawing example is furnished for reference of certain symbols and numbers used. Refer to the numbered notes (shown in a circle on the example) for an explanation of the symbols and numbers.

Example Drawing:



1. Wire numbers are shown on each sheet as **two** or **three** digit numbers (e.g., 01, 02J, 10, 42E, etc) except for wires originating on another sheet, which are shown as up to **seven** digit numbers. In all cases, the complete wire number will appear on the actual wires in the equipment. The first two to four digits indicate the number of the elementary diagram sheet on which the wire originates, and the last two or three digits indicate the wire on that sheet. Thus, 2A04 indicates wire number 04, originating on sheet 2A of the diagram. 100A34J indicates wire number 34J, originating on sheet 100A.

2. This number specifies the point on a specific terminal board (e.g., terminal board 1TB, point 10).

3. These letters show the functional nomenclature of the associated relay.

4. This "cross mapping" number specifies the location of the relay coil controlling the contact shown immediately above. This example shows the TR2 coil is on elementary page 01C, line 17, column M.

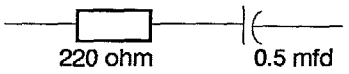
5. This number specifies the wire originating on elementary page 001, wire number 02R.

6. This symbol indicates a panel mounted relay (may be either machine tool relay or time delay relay). Time Delay Relays are defined by the type of timing.

   TDAE - Time delay after energization.

   TDAD - Time delay after de-energization.

7. This symbol indicates a relay coil suppressor as shown below (or similar value) for 115 VAC typical relay coil suppressor:



NOTE: *All relays or contactors added in the proximity of the SCR equipment enclosure or used with control interface logic* ***must*** *have suppressed coils. This includes solenoids, electrical brakes, etc.*

8. This number specifies the location of the relay contact controlled by the coil.. This example shows a time delay contact from the TR2 coil being used on line 09, column H of sheet 1C. Underlined contact locations indicate NC contact.

9. This number [00517F] indicates that this electrical circuit is continued on sheet 005, line 17, column F.

10. This indicates terminal board 4TB, point 46 on the power conversion assembly.

11. This indicates relay K6 on the power conversion assembly PCA01.

12. This indicates a terminal board on the pan assembly; row 2, slot E, point 10.

13. This indicates a card mounted relay in the pan assembly; relay B, on card in row 2, slot C.

14. Pan assembly back plane pin number for wire connection to relay RB-2C, row 2C, pin 17.

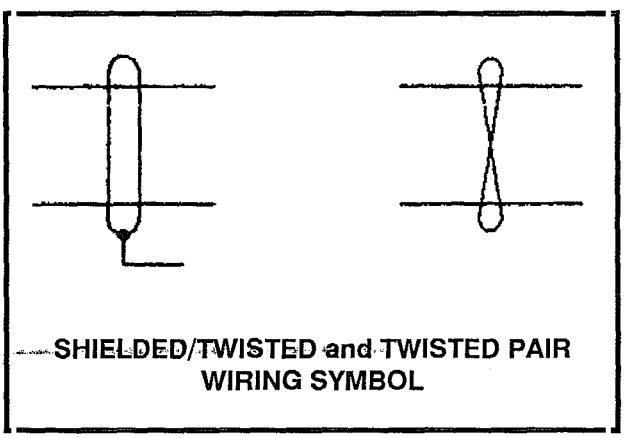
15. Devices and wires shown dashed or in dashed boxes are shown for clarity. Actual devices are located elsewhere in the elementary and are drawn in solid form.

16. This symbol indicates that the associated control wire will be supplied by the customer (see TABLE 2 for complete listing of customer connection symbols).

3.2 GENERAL RULES

1. The following general rules apply to all elementary diagrams unless otherwise shown on the elementary:
   a. All connections to the controller's common must be made to PCOMX.
   b. No connection should be made to +5 VDC.
   c. All digital tach and encoder signals should be differential.
   d. All drive commons (PCOMX) should be grounded at only one point. If the reference is supplied by numerical control or process instrument with a grounded common, the drive common should not be grounded separately.

2. Control system relays, solenoids, or brake coils can produce erratic drive behavior due to electrical noise transients. To reduce this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor connected in series with a 0.5 mfd, 600 volt capacitor can be used. Avoid electromagnetic interference or "noise" introduced by the following:
   a. Radio frequency signals, typically from portable transmitters used in the vicinity of the equipment or its wiring.
   b. Stray high voltage or high frequency signals as might be provided by arc welders.



SHIELDED/TWISTED and TWISTED PAIR WIRING SYMBOL

3. The illustrated symbol (below left) indicates "shielded/twisted" and "twisted pair" wiring. Shielded or Twisted Pair wiring should be used as follows:
   - -- For short wire runs internal to the controller, twisted pair is adequate.
   - -- For wire runs external to the controller, and internal runs longer than 20 feet, shielded/twisted wire is recommended.
   - -- All shield drains should be terminated on one end only, that end being at the controller. This point may be depicted on the elementaries as ground. The remote end of the shield drain wire should be cut off and the wire taped to prevent accidental grounding.
   - -- No shield drain wires should ever be routed through any controller mounted ferrite cores.

4. This equipment may contain thermoswitches or other devices that indicate conditions hazardous to continued equipment operation. It is recommended that the customer connect these devices in stop or alarm circuits in such a manner that prevents accidental restart of the equipment upon resetting of the device.

5. Closed pushbuttons and/or open interlocks indicate maintained circuit closure required.

6. Open pushbuttons indicate momentary circuit closure required.

7. Some single terminal board connection points may be shown more than once for drawing clarity.

8. Conversion unit and card mounted relay contact ratings are 1 amp at 115 VAC or 28 VDC unless otherwise noted.

9. Some additional considerations are as follows:
   - -- Attention should be given to the National Electrical Code and any applicable local codes when installing any drive system. Wire size and insulation type, conduit sizing, enclosures, etc. should be determined per these codes.
   - -- Environments that include excessive amounts of one or more of the following characteristics should be considered hostile to drive system performance and life:
     1. Dust, dirt, or foreign matter.
     2. Vibration or shock.
     3. Moisture or vapors.
     4. Rapid temperature excursions or high ambient temperatures.
     5. Caustic fumes.
     6. Power line fluctuations.
     7. Welding on drive system equipment while control power is on.

4. CUSTOMER WIRING INSTRUCTIONS

4.1 SPECIAL SYMBOLS

Special symbols have been created to clearly show customer connections to the drive control panels, remote operator stations, and other associated devices. These symbols are shown on the elementary diagram whenever a customer connection is required. These symbols are shown below in TABLE 1 and TABLE 2. Typical examples showing the use of these symbols in the elementary are also shown in this section.

TABLE 1: DEVICE LOCATION IDENTIFICATION SYMBOLS

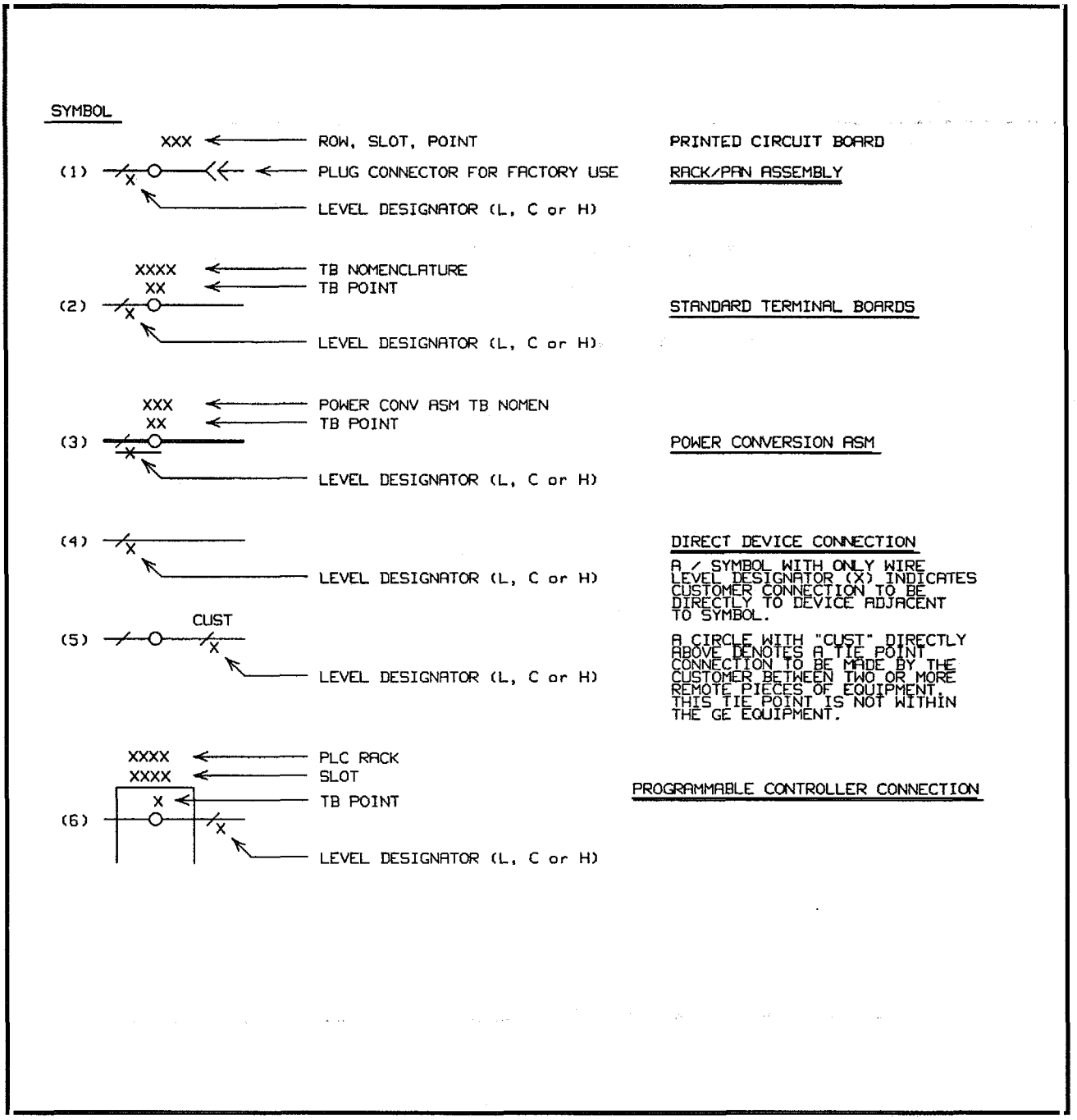
SYMBOL	MOUNTING LOCATION OF COMPONENT
O	Device or circuitry provided by others*.
E	Existing device or component on customer equipment.
R	Device supplied by GE, but remote mounted by customer.
OSM or OS	Main Operator Stations.
OS1, OS2, Etc.	Other Operator Stations.
1TB	Control Terminal Board (high and/or low level connections as defined by drawings).

NOTE: *Other symbols may be specified and defined on the elementary diagrams supplied with the equipment.*

***DISCLAIMER:** The information appearing on this document (GE Drawings) indicated by this symbol was obtained by GE from a third party and is provided solely as a convenience. GE has not independently verified such information and, therefore, does not warrant its accuracy or safe operation.*

TABLE 2: CUSTOMER TERMINAL BOARD CONNECTION SYMBOLS

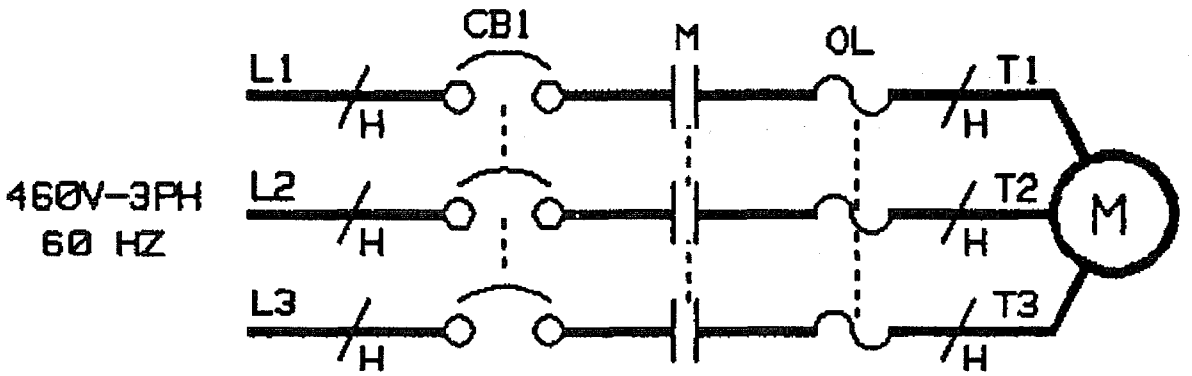
NOTE: *A slash mark (/) is used in the elementary to identify a customer connection point. The following symbols appear on the elementary drawing. Each customer terminal board connection point has the TB nomenclature, the TB point, and the wiring level. Examples shown below are concerned only with customer's external wiring.*



4.2 READING ELEMENTARY EXAMPLES

NOTE: *Certain areas shown in the examples have been circled with dotted lines to purposely emphasize portions of the circuit pertinent to the explanation of customer wiring. Note wire numbers shown on a given sheet of elementary as 01, 22F, L1, P1, etc. are actually prefixed by the sheet numbers 101, 01C22F, 12L1, 14P1, etc. (See Paragraph 3.1).*

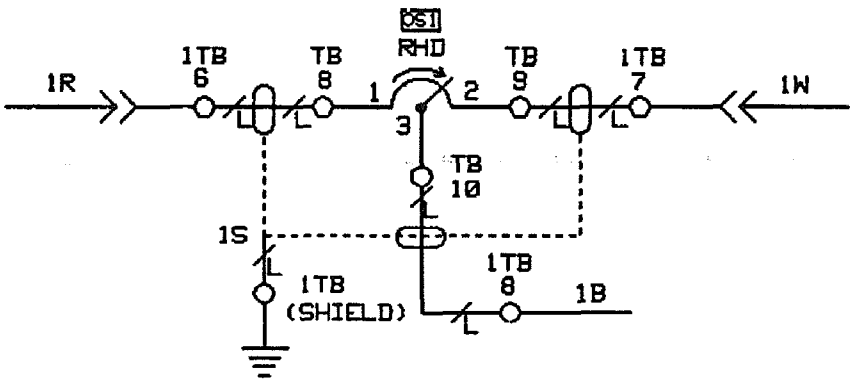
EXAMPLE #1:
Wires L1, L2, L3 are to be run by purchaser from power source directly to the panel mounted circuit breaker. The wire is high level power as indicated by the "H" beneath the symbol.



EXAMPLE #2:
This example assumes operator station (OS1) is supplied and wired by GE. Rheostat "RHD", mounted in OS1, is prewired by wires 1R, 1W, and 1B that run respectively from "RHD" on points 1, 2, 3, to terminal board "TB" in (OS1), points 8, 9, and 10.

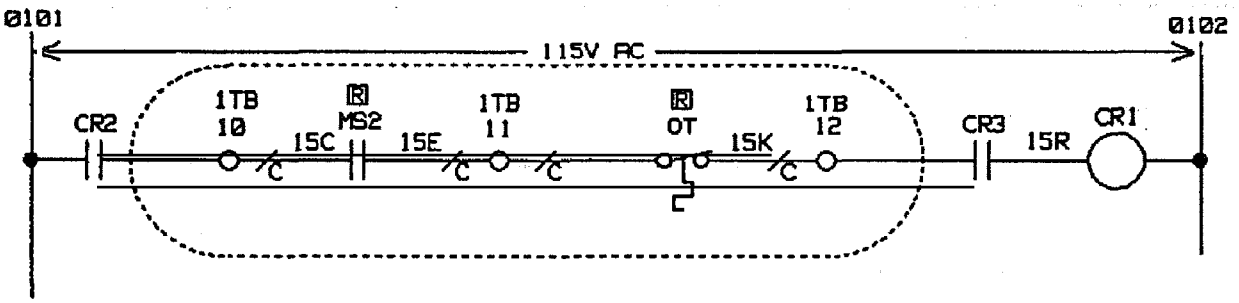
Similarly, wires 1R, 1W, and 1B have been prewired in the control panel to terminal board "1TB" on points 6, 7, and 8. To complete the interconnecting wiring, the customer connects wires 1R, 1W , and 1B from "1TB" points 6, 7, and 8 to operator station "TB" points 8, 9, and 10 respectively. The shield is to be tied to the shield bus on the main control panel.

Wiring by purchaser is a low level (analog or digital) run as defined by the level wiring instructions (i.e., level "L" noted by the L next to the circle). The recommended wire is a 3-conductor shielded wire, and the position of the shielding symbol on the elementary indicates the shielded wire is between the operator station and the main control panel only, and connected to chassis at the main control panel only.



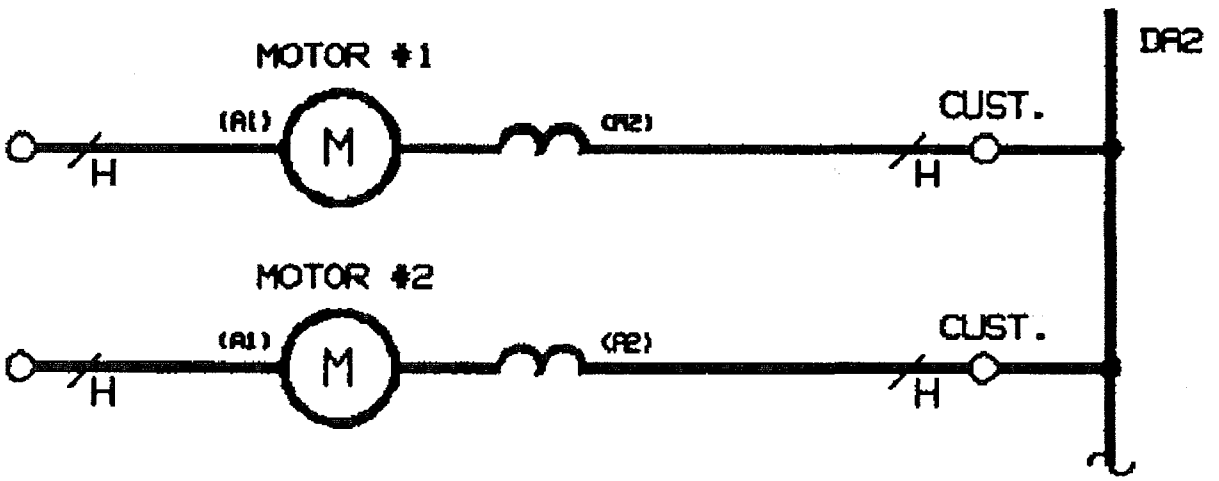
EXAMPLE #3:

In this example, the circle terminal board symbol and the **R** (boxed) symbols over the remote mounted contacts identify the interconnecting wires to be run. Wires 15C and 15K have been pre-connected internally in the control panel to the terminal board "1TB" on points 10 and 12 respectively. "1TB" with wire 15E on point 11 has been reserved as a tie point (customer's option) between the two remote mounted devices. Wires to be run by the customer are wires 15C and 15E from remote contact "MS2" to the control panels "1TB", points 10 and 11. Similarly, wires 15C and 015E from remote contact "MS2" to the control panels "1TB", points 11 and 12 respectively. This wiring by the customer is control level as noted by the **C** next to the slash marks.



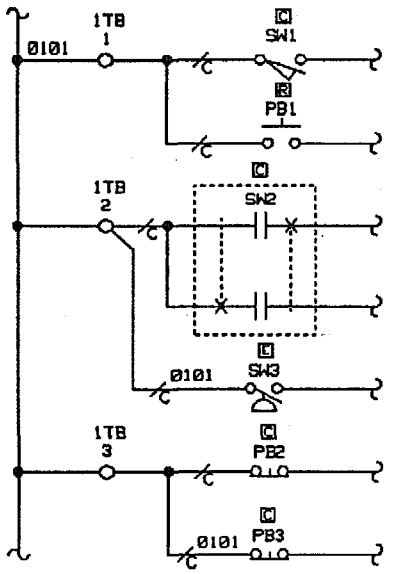
EXAMPLE #4:

In this example, a tie point must be made by the customer between motor #1 and motor #2 (wire DA2). This point is **not** within the GE equipment



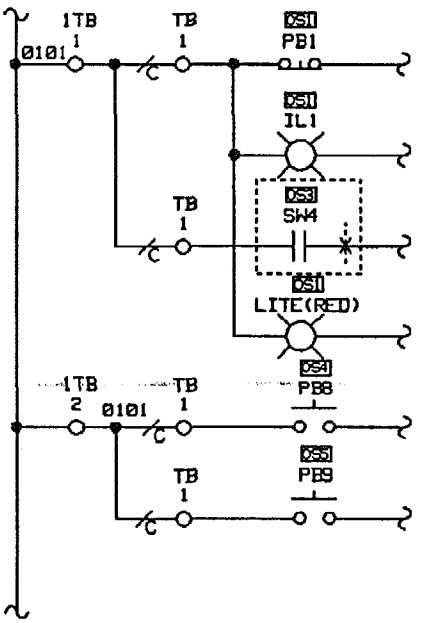
EXAMPLE #5:

This example assumes remote devices (4) are to be mounted by the customer. Provision has been made for the customer to connect two wires from remote devices to the main panel 1TB on point 1, two wires from remote devices to 1TB point 2, and two wires to 1TB point 3, etc. 1TB points 1, 2, and 3 would be jumpered together on factory wired side of the terminal board by GE.



EXAMPLE #6:

In this example, the devices are mounted in operator stations that are supplied by GE and identified as OS1, OS3, 0S4, OS5 (shown boxed). If purchased, all devices within the given operator stations receiving wire 0101 would be wired by GE and taken to terminal boards within the operator stations identified in the example as TB. Provision has been made for the customer to connect two wires to the main control panel 1TB on point 1. One wire would come from operator station OS1, TB point 1 for example, and the other wire from possibly operator station OS3, TB point 1.



5. GENERAL PRACTICES FOR USE WITH DRIVE SYSTEMS PANELS

5.1 WIRING LEVEL DEFINITION AND APPLICATION

In order to guide construction firms in the proper implementation of interconnection wiring, GE has established wiring level separation rules. The main reason for the separation of various levels of wiring is to reduce the effects of electrical noise.

Customer wire runs will carry a level designation. All cables of like levels may be grouped together in trays and conduit. In general, unlike levels must run in separate trays and conduits. Intermixing cannot be allowed, except as defined. Three basic levels are used. These levels are listed below with typical signals defined in each level.

Early planning for level separation is essential to ensure a noise free installation. The level rules should be distributed by the customer's representative to all of the involved electrical and mechanical contractors, as well as the construction personnel. This should be done in all cases to provide adequate awareness of the GE recommendations.

Early consideration of the level rules will also allow the customer to properly plan adequate separation of embedded conduits throughout the process. **On new installations**, sufficient space should be allowed to provide a good and efficient arrangement of mechanical and electrical equipment. **On revamps**, proper consideration of the level rules in the planning stages will enable the customer to make proper application of the level information supplied by GE to provide a more trouble free installation.

The responsibility for application of the level rules defined by GE remains with the customer.

High Level = H

1. AC and DC Switching Circuits
2. AC and DC Busses of 150 Through 700 Volts
3. Motor Armature and Field Circuits
4. 220, 460 VAC Bus Circuits

Control Level = C

1. 105 Through 250 VDC Relay and Contactor Coil Circuits
2. 110, 220 VAC Relay and Starter Coil Circuits
3. Motor Armature Line Shunt Signals for Indication
4. 28 Volt Light and Relay Circuits

Low Level = L

Analog Signals With Less Than 28 VAC Ripple
Digital Signals of 0 - 15 VDC

1. ACOM, DCOM, COM, CCOM, PCOM (Signal Commons)
2. Control Common Tie
3. DC Busses Supplying Sensitive Analog Hardware (+/-5V, +/-10V, +/-15V, +/-20V, +/-24V)
4. Operational Amplifier Signals
5. Power Amplifier Signals
6. Output of Isolation Amplifiers Connected to Sensitive Analog Circuits
7. Analog Tachs and Pilots
8. Phone Circuits
9. DC Busses Supplying Sensitive Digital Circuits and All Associated Wiring With Digital Circuits

Special Level

(**LS** = Low Special, **HS** = High Special)

1. Digital Pulse Tach Signals and Encoders - (LS).
2. Sensitive Analog Circuits Less Than 50mv - (LS).
   Examples: Strain Gages, Thermocouples, Photo Electric Detector Signals, etc.
3. Signals From Line Shunts to "Regulators" - (HS).
4. LAN (Local Area Network) and Communication Signals.

5.2 TRAY AND CONDUIT SPACING

Between levels, 18 inches is the **minimum** distance between the top of one tray and the bottom of the tray above it, or between the sides of adjacent trays. This is also the minimum distance between trays and power equipment of less than 100 KVA.

Between levels, 12 inches is the **minimum** distance between the outside surfaces of adjacent trays and conduits. This is also the minimum distance between trays or conduits and power equipment of less than 100 KVA.

Between levels, 9 inches is the **minimum** distance between the outside surfaces of adjacent conduits being run in banks. This is also the minimum distance between conduits and power equipment of less than 100 KVA.

Tray and Conduit Notes:

1. When unlike signal levels must cross (either in trays or conduits), they should cross at 90 degree angles and at maximum spacing. Where it is not possible to maintain maximum spacing, a grounded steel barrier should be placed between unlike levels at the crossover point.

2. Trays for all levels should be metal, solidly grounded with good ground continuity. Conduits should also be metal to provide shielding. Trays containing low level wiring should have solid bottoms. Ventilation slots or louvers may be used on other trays. Tray covers on low level trays must be used to provide complete shielding. Cover contact to side rails must be positive and continuous to avoid high reluctance air gaps which impair shielding.

3. Trays and conduits should not be routed parallel to high power equipment enclosures of 100 KVA or greater, with spacing of less than five feet for trays and two and one-half feet for conduit.

4. Where practical, the complete power circuit between equipments should be routed in the same path, i.e., the same tray or conduit. This practice will minimize the possibility of power and control circuits encircling each other.

5. When entering equipment, and the spacings previously listed are difficult to maintain, parallel runs should be kept to a minimum and should not exceed five feet in the overall run.

6. **Responsibility for mechanical and environmental locations of conduits, trays, and cables remains with the customer.**

7. Locate low level trays and conduits closest to the control panels.

5.3 SPECIFIC ROUTING OF LEVELS

The following specific practices should be followed to assist in attaining a noise free installation.

Pullboxes and Junction Boxes

1. Within the confines of pullboxes and junction boxes, levels should be kept separate and grounded barriers should be used to assist level spacing.
2. Tray to conduit transition spacings and separations are a potential source of noise. Care should be taken to cross unlike levels at right angles and maintain required separation. Transition areas should be protected in accordance with the level recommendations.

Revamps

1. The reduction of noise on revamps requires careful planning so that low and high levels never encircle each other or run parallel for long distances.
2. The use of existing conduits or trays is practical as long as the level spacing can be maintained for the full length of the run.
3. Where possible, low level cables should be routed in a path apart from existing cables, as these existing cables are generally high level and noise producing.
4. Barriers should be used in existing pullboxes and junction boxes for low level wiring so that the possibility of noise pickup is minimized.
5. Care should be taken so that low level signals are not looped around high level control or power wires in conduits or trays.

Conduits Around and Through Mill or Machinery Housings

1. Care should be taken to plan level spacing on both embedded and exposed conduit in and around the machine or mill. Runs containing mixed levels should be minimized to five feet or less in the overall run.
2. Conduits running through and attached to mill or machinery housing should follow level spacing recommendations and should be discussed with the contractor early in the project.
3. Trunions entering operator's cabinets should be kept as short as possible, thereby minimizing parallel runs or unlike levels. Cables of unlike levels should be kept separate for as long a distance as possible, five feet maximum allowable before entering equipment.
4. Where different levels are running together for short distances, each level should be connected by cord ties, barriers, or some logical method so that intermixing of levels is avoided.

Transitional Areas

1. When entering or leaving conduits or trays, care should be taken to assure that cables of unlike levels do not become intermixed.
2. The use of grounded steel barriers for level separation may be advisable when parallel runs over five feet become necessary.

5.4 COMMON GROUNDING AND POWER SUPPLIES

Many GE control panels utilize an electrical common system which is linked between certain panel lineups to form the common system to which the regulator and logic voltage are referenced. The system must be carried to a customer's solid earth ground from one point only. GE will define the common ground system and to which panel the customer's earth ground will be connected.

The customer should provide a low resistance, separate isolated ground for the panel common system. This ground should not be a part of the normal panel frame or machinery ground system and should be at least 10 feet from any other ground system connections. Acceptable ground sources include grounded metal building columns, ground rods, and ground grid systems.

6. DESIGN PHILOSOPHY OF ELECTRICAL TERMINATIONS

The following practices are used in arrangement of the terminal boards and should be noted when arranging cables.

6.1 GENERAL

1. **Customer supplied equipment** - Only those connections required for GE equipment, including defined tie points or as purchased, will be provided. Customer will be responsible for all other connections.

2. **No more than two wires** should be connected to any one control terminal board point.

6.2 CONTROL PANELS

1. Limit Switches - Limit switch connections will be run directly from the associated control panel. If part of the switch is used in a control circuit foreign to the main control, that portion will be run directly from that equipment and will be wired back to the main control panel. If the foreign circuit is in the same lineup, the wires may be run to the drive functionally defining the limit switch as shown in the elementary. When limit switches or other mill or machine type functions with only one conduit access require the entrance of different levels, arrangements should be made so that the levels are only combined close to the entrance to the device.

2. Selsyns - All selsyn feeders will be run from the control feeder panel to the receiver and on to the transmitter as high level special (HS). Rotor leads will not be on the control panel unless required on that panel.

3. Klixons and BTRs - All klixons and BTRs on a common shaft may be connected at the machine and the two wires brought back to the control panel.

4. Photoelectrc System or Hot Metal Detectors - If the amplifier is separately mounted, only those wires required in GE circuitry will be on the terminal board. If the amplifier is mounted in GE equipment, the terminal boards will contain the wires going to the detector head.

5. Tensiometer Equipment - Only those wires required in GE panels will be on the terminal board.

6. DC Analog Tach-Generator - The control panel terminal board will be used as the tie point for all connections to the tach-generator. Wiring may be run as shielded, twisted pairs in low level. Refer to specific elementary.

7. Pulse Tachometer - Unless otherwise specified in pulse tachometer instruction book or in elementary drawing, wiring from pulse tachometers shall be a direct route to the control panel and through that balun assembly on the conversion unit. Cable specifications are as follows:

   Signal Leads - Each signal lead or lead pair is to be individually shielded with a maximum capacitance conductor to grounded shield of 0.03mfd and minimum conductor size of 22 AWG. Maximum cable length is 500 feet.

   Excitation Power Leads - Power leads (i.e., +12 VDC, 24 VDC, and COM) are to have a maximum resistance of 0.6 ohms each or 1.2 ohms total resistance in a power circuit. Power leads need not be shielded but must be run as low level wiring in the same conduit as shielded signal leads.

8. Communication Cables (Twisted, Shielded, and Balun) - The cable should be as short as possible and not exceed the following **maximum** lengths:

RS232C	50 Feet (15 Meters)
RS422	200 Feet (60 Meters)

The following properties must also be considered when selecting any communication cables:

Shield - It is recommended that the cable incorporate an overall shield which is connected to chassis ground at one end only (preferably the drive end).

Gauge - A minimum wire gauge of 24 AWG is recommended, with heavier gauge required for longer runs.

Capacitance - Low capacitance cable should be used which will provide approximately 35 picofarad per foot or less shunt capacitance between conductors (2500 picofarad maximum for total length on RS232C). Transmission cable between the serial communications isolators should be twinaxial 24 AWG (minimum) twisted-pair, with overall shield and a maximum shunt capacitance of 16 picofarad per foot or less. Separate insulated cables should be used for the transmit and receive pairs to allow isolation of the shields.

Pairing - For RS422 four-wire communications, it is critical that the two transmit wires are paired (twisted together) with each other and the two receive wires are paired with each other. If a transmit wire gets paired with a receive wire, there may be crosstalk which would severely degrade communications reliability.

Ground - A good, low impedance ground conductor of ample size wire must be provided to assure that the drive controller and data equipment grounds are within a few volts of each other or communication errors and equipment damage may result.

Other - Customer must also give specific consideration to mechanical aspects of cables that flex, bend, or move (ie., festoon cables).

NOTE: *Proper cable selection is critical to ensure successful operation of the bus. The maximum length of cable connected to form one communication link depends upon the type of cable used.* ***Mixing different cable types of different terminating resistances on the same bus is not permitted;*** *only one type can be used for all connections on one bus.*

Recommended Wire sizing For LAN (Local Area Network):

LAN - 300 Feet (90 Meters)
Using 531X139AA thru AJ Version Application Card

Isolated LAN - See list below
Using 531X139AK and Later Application Cards

NOTE: *Based on operation at 57.6 baud, wire temperature 50 degrees C, circuitry 0 to 70 degrees C, 0.3 amp current source, H11N2 reciever circuit with maximum of 32 drops.*

ISOLATED LAN WIRING	
Wire Size*	**Maximum Distance From LAN Current Source To Any Receiver****
18 Gauge	600 Feet
16 Gauge	900 Feet
14 Gauge	1400 Feet
12 Gauge	2200 Feet

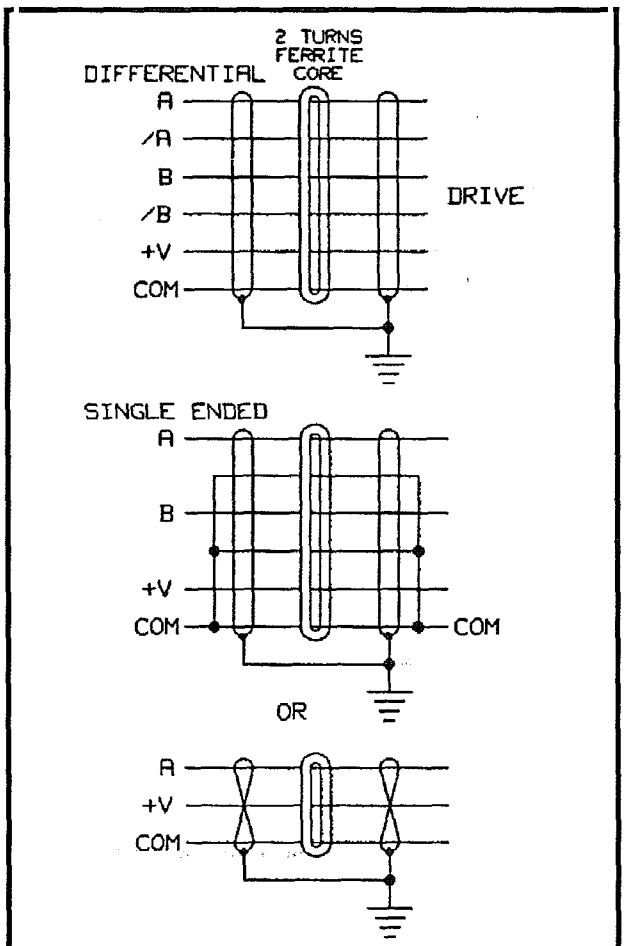
The total capacitance from the wiring on the system must be considered separately from the wire size due to wide variations from different manufacturers. It might be possible to buy wire that is optimized for communications that would have a very low capacitance per foot rating. In any case, for consistent operation at 57.6 baud, the maximum total capacitance tolerable on the isolated LAN from the sum of all wires should be held to* ***less than 0.14 microfarad* *for repeatable and consistent operation. The link, when tested, may tolerate more, but* ***operation with greater than 0.14 microfarad is not gauranteed*** *(number has been adjusted for all known tolerances of system under the worst case operating conditions).*

***Based on wire resistance independent of number of drops.*

9. Panel Wires - Group all wires from one panel to any one specific location together in multiconductor cable providing they are the same level and same type signal, except for special shielded applications as previously described.

6.3 OPERATOR STATIONS

All wiring runs from the control panel directly to the operator station, except those to other control equipments, switchgear, process computers, outside control vendors, tensiometers, or motor control center. Wiring normally runs direct to those functions from the operator station and will not run to the control panel as a tie point. Wires between remote operator stations may return to the control panel.



WIRING EXAMPLES

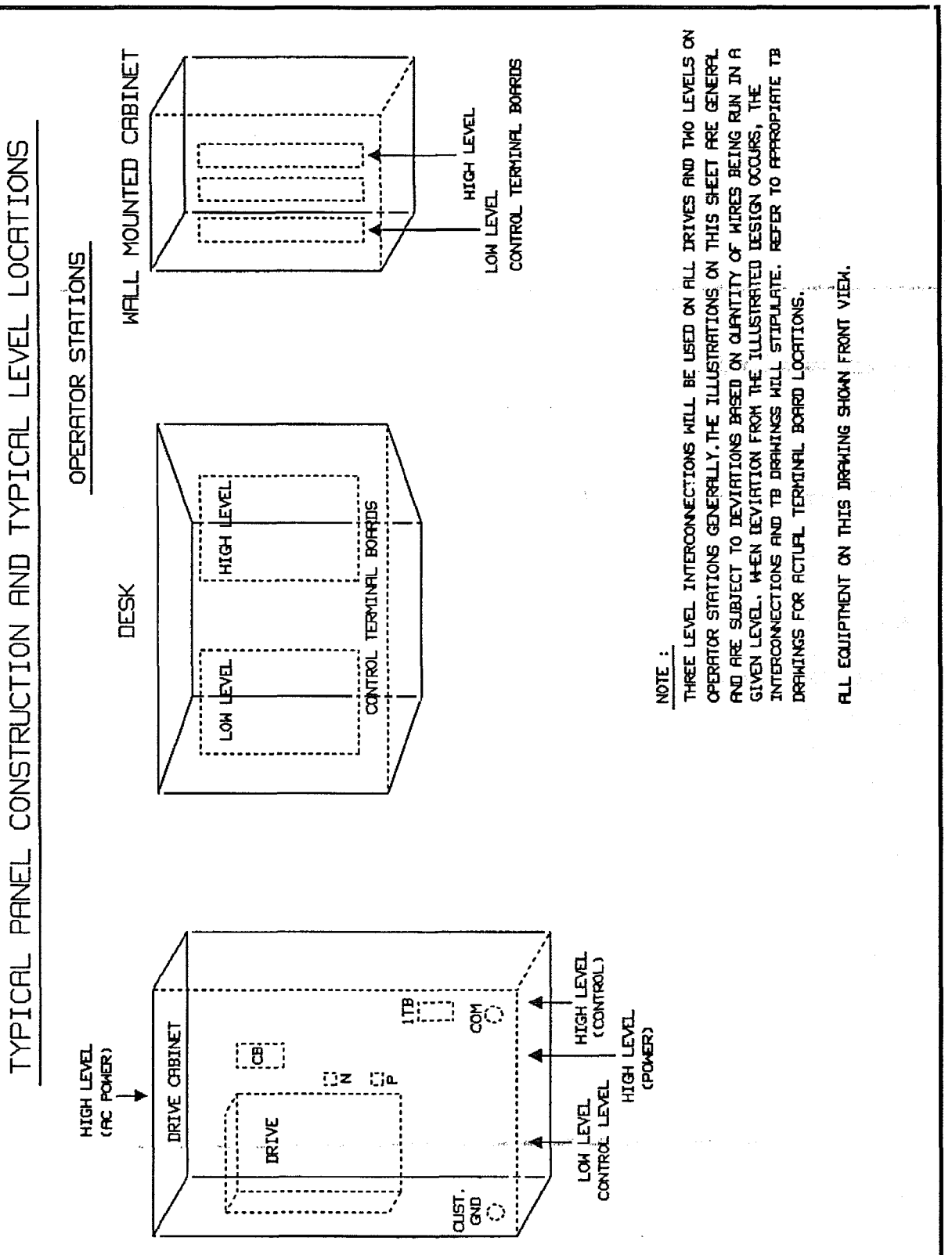


Figure 1. TYPICAL PANEL CONSTRUCTION AND TYPICAL LEVEL LOCATIONS

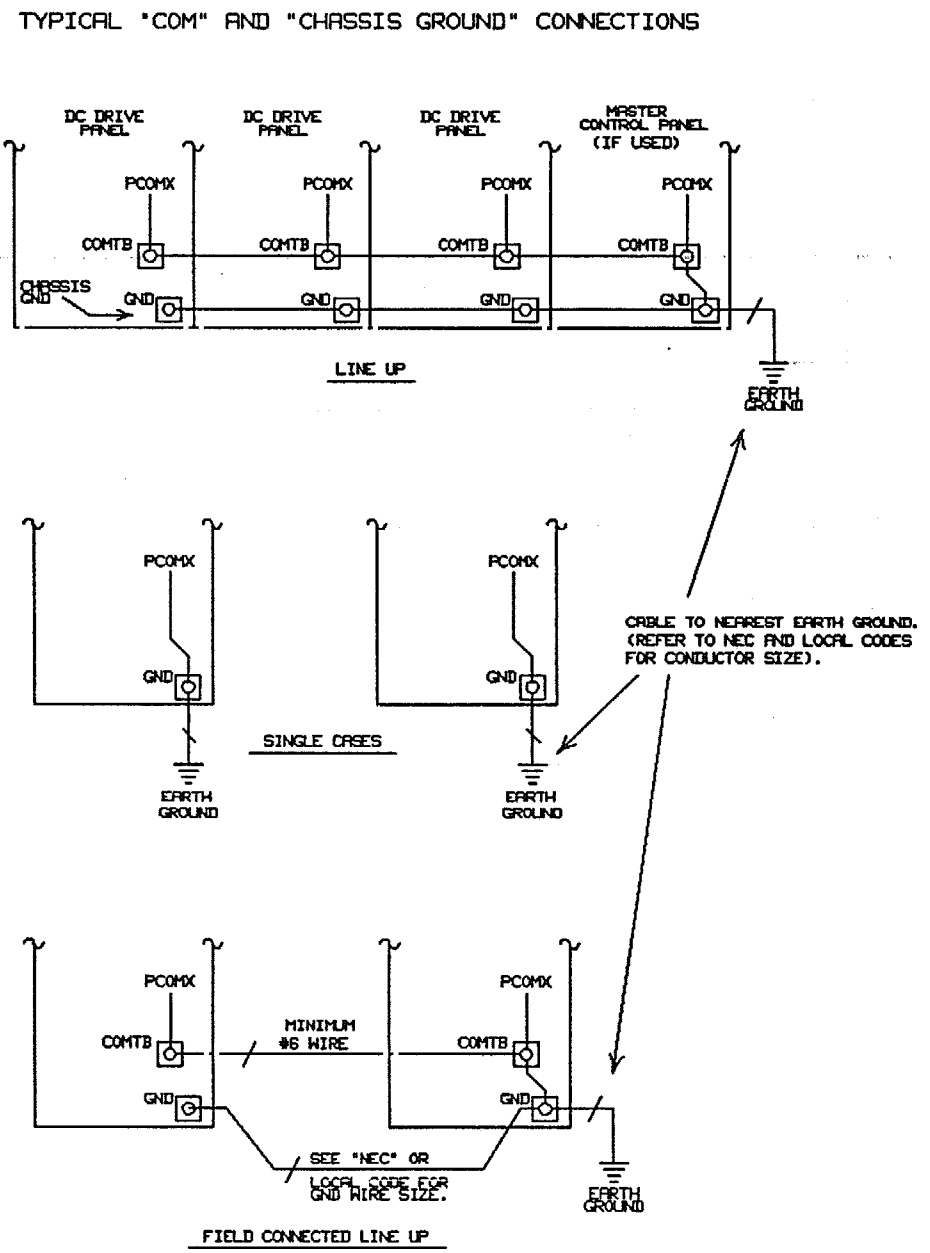


Figure 2. CONTROL COMMON TIES